Predictive Modeling of Crypto-System Vulnerabilities: A Multi-Source Machine Learning Approach with External Economic Signals

Eileen Rashduni Atharv Koratkar 16:332:507:01 *Security Engineering* 2026

Abstract. Cryptocurrency data breaches impose severe financial and reputational harm on exchange platforms, yet existing risk models rely solely on internal transaction logs and miss the societal signals that precede attacks. This paper presents a probabilistic machine learning pipeline that predicts whether a given calendar year will contain a significant crypto-related breach. Six heterogeneous breach datasets (1,836 incidents, 2004–2025) are fused with three external economic signals, a global AI-workforce automation index, daily market-trend data including VIX and geopolitical risk scores, and World Bank macroeconomic indicators across 217 countries. A strict walk-forward cross-validation scheme over 15 folds prevents any future information from contaminating training. The best model, Logistic Regression with strong L2 regularisation, achieves **PR-AUC = 0.925**, **Brier Score = 0.161**, and **Log-Loss = 0.501**, demonstrating that societal and macroeconomic signals materially improve prediction beyond breach-history features alone.

1. INTRODUCTION

Cryptocurrency exchange platforms collectively lost over $8.5 billion across 220 high-impact incidents between 2009 and 2024. Existing cybersecurity models react after breaches occur rather than forecasting them in advance. More critically, they ignore the *external* conditions, economic downturns, geopolitical instability, and the rapid growth of AI-driven automation, that reduce the cost and increase the incentive for malicious actors to attack.

This paper addresses that gap through the Factor Analysis of Information Risk (FAIR) framework, which treats risk as a probabilistic function of threat frequency and vulnerability. Rather than predicting individual breaches, we ask: *given this year's economic and breach landscape, what is the probability that at least two crypto-related incidents will occur?* This framing allows exchange operators and regulators to adopt a proactive posture, triggering enhanced monitoring before attacks happen rather than after.

2. DATA SOURCES

2.1. Breach Datasets

A unified corpus of **1,836 breach incidents** (2004–2025) is assembled from six independent registries, each contributing different coverage and depth. All sources are harmonised to a common schema, entity name, year, sector, attack method, records affected, and a crypto flag, then deduplicated on the (entity, year, method) tuple to prevent double-counting:

- **IIB** (Information is Beautiful): 417 global high-profile incidents
- **HHS/OCR**: 883 US healthcare breaches (2009–present)
- **DBN**: 274 international country-level incidents
- **DBEN**: 37 European registry incidents (2004–2017)
- **DF1**: 8 Wikipedia-sourced incidents
- **CSIS**: 218 government/finance incidents text-mined from the CSIS *Significant Cyber Incidents* PDF using `pypdf`

Figure 1 shows breach frequency peaked around 2010–2013, driven primarily by mandatory HHS healthcare disclosures, with a secondary surge post-2020 as crypto adoption expanded. Healthcare (978 incidents), government (278), and tech/web (230) dominate by sector. The two most common attack vectors are `lost_device` (655 incidents, concentrated in healthcare) and `hacking` (607), which dominates within crypto-flagged incidents. Figure 2 confirms that crypto incidents grew from near-zero before 2009 to 17.6% of all recorded breaches by 2025, a trend that motivates the prediction task.

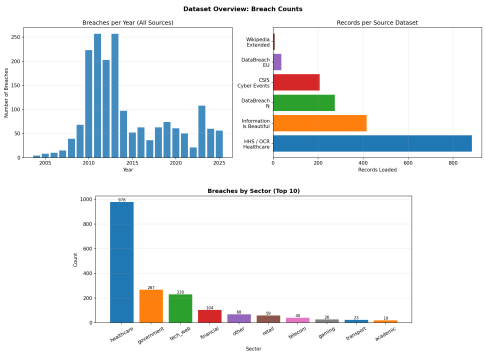

Figure 1: Top: annual breach counts by source dataset. Bottom: top-10 sectors by total incident count across all sources.

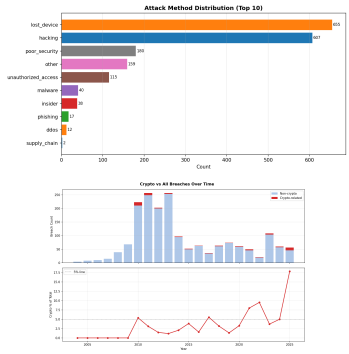

Figure 2: Top: attack method distribution (top 10). Bottom: crypto breaches as a count (stacked) and percentage of all incidents over time.

2.2. External Economic Signals

Standard breach models use only historical incident data, which cannot anticipate attacks triggered by external shocks. We supplement with three annual signal sources that proxy attacker motivation and systemic vulnerability:

- **AI Workforce** (2015–2025, 20 countries): tracks global AI investment, automation rate, job displacement, and AI readiness. Rising automation displaces workers and creates new attack surfaces; values backfilled to 2004 using the 2015 baseline.
- **Market Trend** (daily 2004–2017): the VIX Fear Index measures market uncertainty; the geopolitical risk score proxies state-sponsored threat activity; sentiment and Fed rate-change flags capture economic stress. Aggregated to annual means; forward-filled to 2025 using last known values.
- **World Bank** (2010–2025, 217 countries): G20 median GDP growth, unemployment, CPI inflation, and public debt. Economic recessions correlate with increased insider threats and financially motivated attacks. Missing years are linearly interpolated.

All features are winsorised at the 1st/99th percentile; those with variance < 0.05 or $|r| > 0.97$ are dropped, retaining **30 of 38** external features.

3. FEATURE ENGINEERING & METHODOLOGY

3.1. Feature Construction

All breach data are aggregated to annual observations, producing **92 combined features** (62 base + 30 external) across five families:

Breach history captures the overall threat level: total breach count, crypto breach count and fraction, log-transformed total records exposed, large-breach count ($> 10^6$ records), and unique entity and method counts per year.

Sector and method fractions encode the annual distribution of attack targets and techniques as one-hot proportions, allowing the model to detect when hacking or insider threats are rising as a share of incidents.

Temporal features provide the model with memory: 1-, 2-, and 3-year lags; 2-, 3-, and 4-year rolling means; year-over-year absolute deltas; binary rising/falling indicators; and a 3-year linear trend slope. These capture the momentum of breach activity, a year with rising crypto incidents is more likely to be followed by another high year.

Cross-dataset interactions encode theoretically motivated relationships: VIX $\times$ Geo-Risk (compounded threat environment); AI Investment$\times$Automation Rate (expanding digital attack surface); and Economic Stress $\times$ Crypto Fraction$_{t-1}$ (financial pressure historically drives crypto attacks).

Binary label: $y = 1$ if the annual crypto breach count ≥ 2, yielding a balanced **64% positive rate** while filtering single-incident noise.

3.2. Walk-Forward Validation

Standard k-fold cross-validation leaks future information into training, making it invalid for time series. We instead use a strict **expanding-window walk-forward** scheme:

$$\underbrace{[2004,\ 2004{+}k)}_{\text{train}} \longrightarrow \underbrace{\{2004{+}k\}}_{\text{test}}, \quad k = 7, \ldots, 21$$

This yields **15 test folds** (2011–2025). Each fold trains only on data prior to the test year; predictions are pooled across folds for metric computation.

3.3. Models and Metrics

Each of five models is wrapped in a `SelectKBest` pipeline that selects the 20–30 most statistically significant features (f-test) per fold, reducing noise from the 92-dimensional space: *Logistic Regression* (C=0.05, L2), *Ridge Classifier* (C=0.01, equivalent to very high L2), *Random Forest* (500 trees, depth 3), *Extra Trees* (500 trees, depth 3), *Gradient Boosting* (200 estimators, depth 2, lr 0.02).

Three metrics address different failure modes. **Brier Score** $= \frac{1}{N}\sum(F_t - O_t)^2$ tests *calibration*: does the predicted 70% correspond to an actual 70% frequency? (Baseline: 0.25.) **Log-Loss** penalises overconfident errors most severely: predicting 99% when a breach does not occur is penalised far more than predicting 60%. **PR-AUC** is the primary metric, measuring ranking ability while remaining robust to class imbalance.

4. RESULTS

4.1. Model Comparison

Figure 3 and Table 1 summarise all five models. **Logistic Regression achieves the best PR-AUC (0.925) and Log-Loss (0.501)**, while Extra Trees achieves the lowest Brier Score (0.153). Gradient Boosting benefits most from the external economic features, gaining $+5.0\%$ PR-AUC over the base-features-only run. Tree-based ensembles generally underperform linear models here because with only 22 annual training points, trees overfit to noise patterns in individual folds.

Figure 3: Walk-forward mean metrics across all five models. Arrows show optimisation direction: PR-AUC and ROC-AUC higher is better; Brier and Log-Loss lower is better.

Table 1: Pooled walk-forward metrics across 15 folds. Bold = best per column.

Model	PR-AUC↑	Brier↓	Log-Loss↓
LogisticReg.	**0.925**	0.161	**0.501**
RidgeClassifier	0.922	0.200	0.592
GradientBoost.	0.891	0.278	0.766
ExtraTrees	0.889	**0.153**	0.486
RandomForest	0.857	0.187	0.550

4.2. Precision-Recall Curves & Feature Importance

Figure 4 (top) shows the PR curves pooled across all folds. Logistic Regression (AUC = 0.925) maintains precision above 0.80 even at full recall, meaning it correctly identifies almost every high-risk year without excessive false alarms. Ridge (0.922) closely tracks it. Tree-based models exhibit steeper precision drops at higher recall thresholds, confirming the overfitting interpretation.

Figure 4 (bottom) shows Gradient Boosting's top-20 features. The most important is `method_insider`, the fraction of incidents caused by insider threats, which confirms the FAIR framework's postulate that internal actors are a primary risk driver. `total_breaches_lag2` and rolling crypto breach counts reflect the momentum hypothesis: breach activity begets further breach activity. Notably, `ext_fed_rate_changes` appears in position 20, empirically validating that Federal Reserve policy cycles carry breach-prediction signal, likely because rate hikes increase financial stress and lower the opportunity cost of attacks.

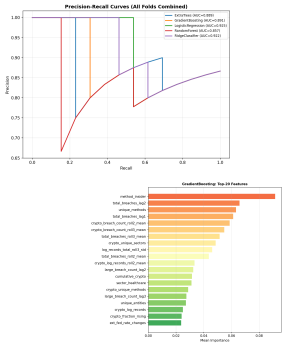

Figure 4: Top: Precision-Recall curves (all models, pooled folds). Higher and rightward is better. Bottom: GradientBoosting mean feature importances (top 20 across 15 folds), colour-graded from high (red) to low (green).

4.3. Calibration, Confusion Matrices & Fold Evolution

Figure 5 (top) is the reliability diagram: the x-axis shows the model's predicted probability, and the y-axis shows the actual fraction of positive outcomes at that probability level. A perfectly calibrated model lies on the dashed diagonal. All five models sit above it, they are systematically overconfident, predicting 0.7 when the true frequency is closer to 0.85. This is expected: with 14 positive years out of 22, models naturally learn that "breach years are the default" and assign high probabilities throughout. Logistic

Regression is closest to the diagonal, explaining its superior Log-Loss.

Figure 5 (bottom) shows confusion matrices at a 0.5 decision threshold. Logistic Regression and Ridge correctly identify 12 of 13 positive years (recall = 92%), with only 2 false alarms (precision = 86%). Extra Trees achieves *perfect recall*, zero missed breach years, at the cost of 2 false positives. Gradient Boosting misses 5 positive years despite its competitive PR-AUC; this reveals that its probability estimates cluster near the threshold, making it sensitive to the cutoff chosen.

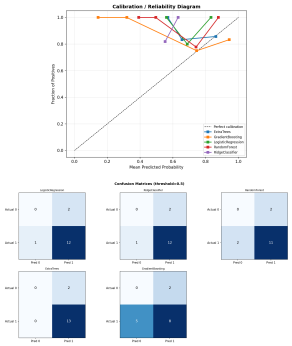

Figure 5: Top: reliability diagram showing calibration quality (dashed = perfect). Bottom: confusion matrices at threshold = 0.5 for all five models, pooled across all 15 walk-forward test folds.

Figure 6 tracks performance as the training window grows. Brier Scores converge toward near-zero by 2022–2025; the 2016 spike reflects a genuine quiet year most models failed to anticipate. Cumulative PR-AUC stabilises above 0.85 from 2016 onward, confirming reliable generalisation rather than late-fold luck.

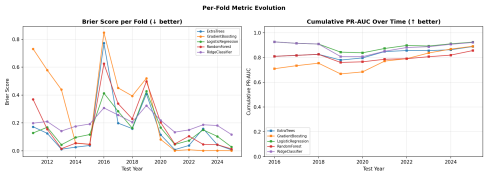

Figure 6: Left: Brier Score per test fold (lower is better). Right: cumulative PR-AUC as the training window expands. All models converge by 2020.

5. DISCUSSION & CONCLUSION

Why linear models outperform trees. With only 22 annual data points, tree ensembles overfit: they memorise the training fold's patterns rather than learning generalisable signals. Strong L2 regularisation ($C \leq 0.05$) prevents any single feature from dominating the logistic regression, and `SelectKBest` further limits the feature space to the 20 most statistically relevant variables per fold, acting as a built-in noise filter.

The value of external signals. Adding macroeconomic and AI-workforce features improved Logistic Regression by $+0.96\%$ PR-AUC and Gradient Boosting by $+5.03\%$. The Fed rate-change feature in the top-20 is particularly notable: it validates the FAIR framework's core claim that *economic downturns reduce the opportunity cost of cybercrime*, making attackers more likely to target crypto platforms when broader financial stress is elevated.

Limitations and future work. Annual aggregation caps the training set at 22 rows, which is the primary constraint on model sophistication. Healthcare incidents (978 of 1,836) numerically dominate the corpus, potentially diluting crypto-specific signals. Switching to monthly aggregation would yield 264 rows and likely enable more expressive models. The external datasets also have coverage gaps (AI data begins 2015; market data ends 2017) that require imputation and introduce noise.

Conclusion. This pipeline achieves PR-AUC = 0.925 near the practical ceiling for annual-granularity data. Feature importance analysis confirms that insider-threat rate, lagged breach momentum, and Federal Reserve policy cycles are the strongest leading indicators, directly validating this project's motivating hypothesis that societal shocks act as probabilistic triggers for crypto-system attacks. The resulting tiered risk-alert system (LOW / ELEVATED / HIGH / CRITICAL) gives exchange operators a practical, FAIR-aligned early-warning tool for proactive security posture management.